C. Kuroni and Impossible Calculation

time limit per test: 1 second
memory limit per test: 256 megabytes

To become the king of Codeforces, Kuroni has to solve the following problem.

He is given n numbers $a_1, a_2, \dots, a_n$. Help Kuroni to calculate $\prod_{1 \le i < j \le n} |a_i - a_j|$. As result can be very big, output it modulo m.

If you are not familiar with short notation, $\prod_{1 \le i < j \le n} |a_i - a_j|$ is equal to $|a_1 - a_2| \cdot |a_1 - a_3| \cdot \dots \cdot |a_1 - a_n| \cdot |a_2 - a_3| \cdot |a_2 - a_4| \cdot \dots \cdot |a_2 - a_n| \cdot \dots \cdot |a_{n-1} - a_n|$. In other words, this is the product of $|a_i - a_j|$ for all $1 \le i < j \le n$.

Input

The first line contains two integers n, m ($2 \le n \le 2 \cdot 10^5$, $1 \le m \le 1000$) — number of numbers and modulo.

The second line contains n integers $a_1, a_2, \dots, a_n$ ($0 \le a_i \le 10^9$).

Output

Output the single number — $\prod_{1 \le i < j \le n} |a_i - a_j| \bmod m$.

Examples

input
```
2 10
8 5
```
output
```
3
```

input
```
3 12
1 4 5
```
output
```
0
```

input
```
3 7
1 4 9
```
output
```
1
```

Note

In the first sample, $|8 - 5| = 3 \equiv 3 \bmod 10$.

In the second sample, $|1 - 4| \cdot |1 - 5| \cdot |4 - 5| = 3 \cdot 4 \cdot 1 = 12 \equiv 0 \bmod 12$.

In the third sample, $|1 - 4| \cdot |1 - 9| \cdot |4 - 9| = 3 \cdot 8 \cdot 5 = 120 \equiv 1 \bmod 7$.